Using Calculators to Sketch Graphs

Exam code: 0607

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Written by: Mark Curtis
Reviewed by: Dan Finlay
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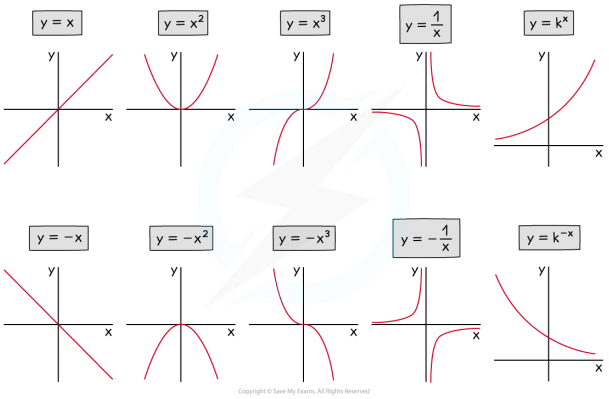
Using Calculators to Sketch Graphs

How do I draw graphs on my graphic display calculator?

- Navigate on your calculator to the **Add Graphs** button
  - You may need to start a new document first
  - On some models you will need to select **graph mode** in the menu
- Enter the **equation** of the function on the line **f(x)=**
  - This is the function, $f(x)$
  - On other models it may be labelled as **f1(x)=** or **Y1:**
- Press **enter** or **EXE** and the graph should appear
  - You may need to go to the menu or press the **zoom** button to zoom in or out
    - The "**zoom – fit**" button (or equivalent) automatically zooms for you on the key features
    - Other models may also have a "**V-Window**" menu where you can select the exact range of values to show

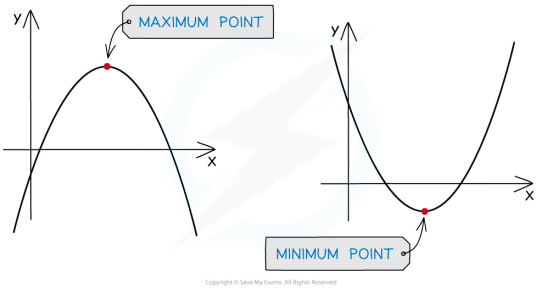


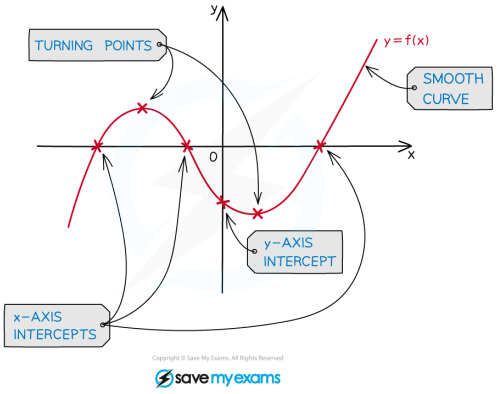
How do I find graph features using my graphic display calculator?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **Zeros (or roots):** these are the ***x*-intercepts**
  - You may have to click before and after the *x*-intercept you require so that the calculator knows **where to search**
  - **y-intercepts** can be found **without a calculator** by substituting $x = 0$ into $f(x)$
    - though some models do include a **Y-ICEPT** button which can be used

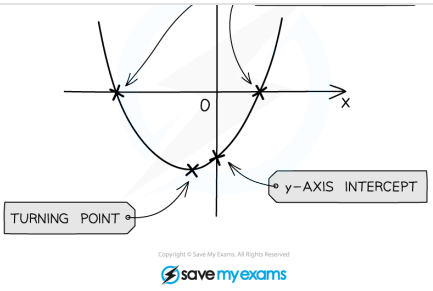
- **Maximum or minimum points:** these are the coordinates of the **highest point** and **lowest point** on the graph
  - You may have to click before and after the point you require so that the calculator knows where to search
  - These can be called **local maxima** or **local minima**
  - They are also referred to as **turning points**
    - or the **vertex** if talking about a **quadratic** graph



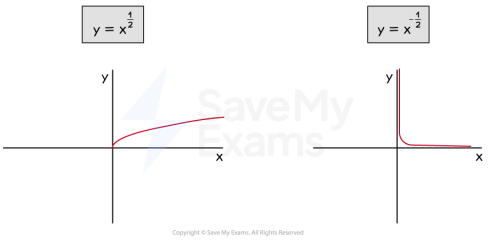


How do I sketch a graph in the exam from my graphic display calculator?

- Exam questions may ask you to **sketch** the graph on a given blank set of axes
  - You do **not** need to **plot** points exactly
    - and you do **not** need a **table** of values
  - **Draw** a **single smooth freehand** curve that copies the shape of the curve on your graphic display calculator
    - Do **not** use a ruler for curves!
    - Copy any **key features** and **symmetries**



- Exam questions may ask you to use your calculator to sketch an **unknown function** that you have not seen before



How do I create a table of values on my graphic display calculator?

- Your calculator can create **tables of values** for you, if needed
- Find the **table** function
  - Type in the graph **equation** (the **function**, f(x)=)
  - Enter the **start** value (or equivalent)
    - The first *x*-value in the table
  - Enter the **end** value (or equivalent)
    - The last *x*-value in the table
  - Enter the **step size** (or equivalent)
    - How big the steps (gaps) are from one *x*-value to the next
  - You may have to select **SET** to set the above values
  - Press **enter** or **EXE** to see the table of values

Examiner Tips and Tricks

- Exam questions will say "draw this graph on your calculator" to make it clear when using a calculator is intended

use it to sketch the curve on the axes provided, showing coordinates of all axes intercepts and any maximum or minimum points.

Start a new graph on your calculator and enter
$f(x) = 2x^2 - 4x - 6$ to generate the graph
This shows the curve has a minimum point (not a maximum point)

Select the option to "analyze" the graph (this may be labelled as G-Solv)
Choose the option "zeros" (or roots) to find the x-intercepts
Choose the option "minimum" (or equivalent) to find the coordinates of the minimum point

The question asks for all axes intercepts, so this includes the y-intercept
Substitute $x = 0$ into $f(x) = 2x^2 - 4x - 6$ to get -6
(or you can use the Y-ICEPT button on some models)

Show all the above information on a sketch using coordinates

(b) Match the following curve equations to the sketches below. You may use your graphic display calculator in this question.

(1) $y = 0.6x + 2$ **(2)** $y = 3^x$ **(3)** $y = -0.7x^3$ **(4)** $y = \frac{4}{x}$ **(5)** $y = -x^2 + 3x + 2$

(1) is a straight line equation ($y = mx + c$) so must match graph **D**

(2) requires plotting on a graphic display calculator to see that it is A

(3) requires plotting on a graphic display calculator to see that it is E

(4) requires plotting on a graphic display calculator to see that it is B

(5) is a quadratic equation with a negative coefficient so matches graph C

When plotting these graphs on your calculator, you may need to use the zoom function or V-Window button to see the overall shapes of the graphs

(1) → D
(2) → A
(3) → E
(4) → B
(5) → C

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